FAZIL KHAN

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Education

JOHNS HOPKINS UNIVERSITY | *Baltimore, MD* — *May 2026*
MSE Robotics

GOVERNMENT COLLEGE OF ENGINEERING | *India* — *June 2023*
B.Tech. Electrical Engineering

Professional And Research Experiences

Perception Researcher | *MIRACLE Lab, Johns Hopkins University | Baltimore, MD* — *March 2026 - Present*

- Constructing a deformable tissue real-to-sim pipeline by coupling stereo perception with position-based dynamics.
- Integrating **self-supervised Monodepth2** and **DeepLabV3+** for depth estimation and semantic segmentation.
- Accelerating tissue simulation via **GPU-parallelized** shape-matching and **SDF-based** registration from point clouds.

Robotics Researcher | *LCSR, Johns Hopkins University | Baltimore, MD* — *July 2025 - February 2026*

- Designed and implemented a real-time **closed-loop** control architecture for **MRI-guided** robotic needle steering, integrating **adaptive Jacobian control** with **state estimation** for improved surgical precision.
- Built **Extended Kalman Filter** for needle tip estimation with **90 microns** error using MRI updates on **30%** of control cycles.
- Developed adaptive Broyden-based Jacobian controller for nonlinear needle-tissue interaction; **93%** success.
- Implemented **controller chaining** in **ROS2 Control**, cutting end-to-end latency by **50%** versus topic-based communication.

Graduate Engineer | *Reliance Industries Limited | India* — *August 2023 - May 2024*

- Built **PatchCore** & **WideResNet50** unsupervised anomaly detection to replace manual inspection of reflective bobbins.
- Mitigated specular occlusion by coupling **cross-polarized** hardware with **Convolutional Block Attention Modules (CBAM**).
- Achieved **98.5% AUROC** and sub-**15ms** inference latency by deploying the computer vision pipeline via **TensorRT.**
- Led an independent initiative integrating **180°** LiDAR into autonomous shuttles, reducing curved-route collisions by **70%.**
- Collaborated on CapEx automation, installing 4 Yaskawa **6-DOF** robots on the bobbin line to boost production speed by **20%**.

Projects

VLM-Guided Mobile Manipulation for Autonomous Object Retrieval | *Prof. Simon Leonard* — *May 2026*

- Designed ROS 2 stack for object retrieval on **TurtleBot4 + OpenManipulator-X**; built in **Gazebo**, validated on real **hardware**.
- Engineered **Gemini Robotics-ER** VLM perception with **text-prompt** detection, depth + **PCA-based** 6-DoF pose, and one-shot latch + 30 Hz TF chain-republish to handle API rate limits and cut latency; benchmarked vs **ArUco PnP** at **~3** cm error.
- Built ROS2 **Nav2 + AMCL** state machine for autonomous target search and visual-servo approach; **84%** planning success rate.
- Implemented **MoveIt 2** pick-and-place action server with **IK**, joint-space + Cartesian planning on **4-DoF** OpenManipulator-X.

Semantic SLAM | *Prof. Kapil D. Katyal* — *December 2024*

- Integrated **ORB-SLAM3** and **YOLOv7** in **ROS** for real-time monocular semantic SLAM and **3D object mapping**.
- Trained YOLOv7 on 1,088 annotated images, achieving **95.7% mAP@.5** and **88.7%** precision across 4 object classes.
- Calibrated a Raspberry Pi camera and built a ROS pipeline for streaming, **depth estimation**, and **RViz** visualization.

Rough Terrain Navigation using Model-Based RL | *Prof. Enrique Mallada* — *November 2025*

- Modeled terrain navigation as an **MDP** to address compounding model uncertainty in divergent long-horizon dynamics.
- Designed a probabilistic CNN-LSTM dynamics model with multistep loss, achieving **4×** better performance than baselines.
- Engineered closed-loop pipeline with **PD tracking** & Augmented Lagrangian optimization for stable non-myopic planning.
- Achieved **82%** success rate in **250 PyBullet** trials, outperforming penalty-based (60%) & agnostic (66%) methods.

Nonlinear MPC for Spacecraft Attitude Control | *Prof. Abhishek Cauligi* — *October 2025*

- Developed quaternion-based **MPC** for NASA Astrobee robot using **acados** with 7-state dynamics and 3-axis torque control.
- Achieved target attitude convergence over **50-step** horizon with bang-bang profiles under **0.03** Nm actuator limits.

Technical Skills

Programming Languages and Dev Tools: Python, C++, MATLAB, CMake, Bash, colcon, Git, CI/CD, Docker, Linux.
Robotics: ROS 2, ros2_control, MoveIt, Nav2, CasADi, acados, KDL,Eigen, tf2, URDF, Gazebo, Ignition, RViz, PyBullet, MuJoCo.
Deep Learning: PyTorch, TensorFlow, Stable-Baselines3, CUDA, YOLO, Monodepth2, DeepLabV3+, OpenCV, JAX, NumPy, SciPy.

Teaching Experience

Graduate Teaching Assistant | EN.601.783 Vision As Bayesian Inference | *Johns Hopkins University* — *January 2026 - May 2026*

- Hosted office hours and graded assignments for Prof. Alan Yuille's graduate-level course of 90+ students.